PREMIUM

Full-Length Mock Test

Mock 02

Electronics and Communication Engineering

GATE EC

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Network Theory — AC Circuits]

Thevenin equivalent replaces any linear network by:

A) V-source + Resistor
B) I-source + Resistor
C) V-source + Inductor
D) I-source + Capacitor

Q.2 [1 Mark] [EASY] [Signals & Systems — Convolution]

Impedance of inductor L in s-domain:

A) 1/s
B) sL
C) 1/(sL)
D) s/L

Q.3 [1 Mark] [EASY] [Control Systems — Block Diagrams]

Routh-Hurwitz determines:

A) Steady-state error
B) RHP pole count
C) Damping ratio
D) Natural frequency

Q.4 [1 Mark] [EASY] [Digital Electronics — Combinational Circuits]

BJT in active region:

A) Both junctions forward
B) Both reverse
C) EB forward, CB reverse
D) EB reverse, CB forward

Q.5 [1 Mark] [EASY] [Analog Electronics — Feedback]

8085 microprocessor address lines:

A) 8
B) 16
C) 20
D) 32

Q.6 [1 Mark] [EASY] [EMFT — Magnetostatics]

FM modulation index ;" v—F, ”f=75kHz, fm=15kHz:

A) 3
B) 5
C) 10
D) 15

Q.7 [1 Mark] [EASY] [Communication — Digital Modulation]

Intrinsic impedance of free space ;r $f

A) 50:•

B) 120< ©

C) 377:•

D) Both B and C

Q.8 [1 Mark] [EASY] [Microprocessors — Programming]

A 4-bit DAC with Vref=5V has resolution:

A) 0.3125V
B) 0.5V
C) 1V
D) 0.25V

Q.9 [1 Mark] [EASY] [Network Theory — Thevenin/Norton]

A JK flip-flop with J=K=1 toggles on:

A) Every clock
B) Only J change
C) Only K change
D) Never

Q.10 [1 Mark] [EASY] [Signals & Systems — LTI Systems]

Binary 1011 to Gray code:

A) 1011
B) 1110
C) 1101
D) 0111

Q.11 [1 Mark] [EASY] [Control Systems — Routh-Hurwitz]

Polarization in EM waves refers to:

A) Electric field direction
B) Magnetic field direction
C) Wave propagation direction
D) Frequency variation

Q.12 [1 Mark] [EASY] [Digital Electronics — Boolean Algebra]

Nyquist rate for signal with max frequency 5kHz:

A) 5 kHz
B) 10 kHz
C) 2.5 kHz
D) 15 kHz

Q.13 [2 Marks] [MODERATE] [Analog Electronics — Feedback]

Thevenin equivalent replaces any linear network by:

A) V-source + Resistor
B) I-source + Resistor
C) V-source + Inductor
D) I-source + Capacitor

Q.14 [2 Marks] [MODERATE] [EMFT — Transmission Lines]

Impedance of inductor L in s-domain:

A) 1/s

B) sL

C) $1/(sL)$

D) s/L

Q.15 [2 Marks] [MODERATE] [Communication — AM/FM]

Routh-Hurwitz determines:

A) Steady-state error
B) RHP pole count
C) Damping ratio
D) Natural frequency

Q.16 [2 Marks] [MODERATE] [Microprocessors — 8085/8086]

BJT in active region:

A) Both junctions forward
B) Both reverse
C) EB forward, CB reverse
D) EB reverse, CB forward

Q.17 [2 Marks] [MODERATE] [Network Theory — Three-phase]

8085 microprocessor address lines:

A) 8
B) 16
C) 20
D) 32

Q.18 [2 Marks] [MODERATE] [Signals & Systems — Laplace Transform]

FM modulation index ;" v—F, ”f=75kHz, fm=15kHz:

A) 3
B) 5
C) 10
D) 15

Q.19 [2 Marks] [MODERATE] [Control Systems — Block Diagrams]

Intrinsic impedance of free space ;r $f

A) 50:•
B) 120< ©
C) 377:•
D) Both B and C

Q.20 [2 Marks] [MODERATE] [Digital Electronics — Combinational Circuits]

A 4-bit DAC with Vref=5V has resolution:

A) 0.3125V
B) 0.5V
C) 1V
D) 0.25V

Q.21 [2 Marks] [MODERATE] [Analog Electronics — Diodes]

A JK flip-flop with J=K=1 toggles on:

A) Every clock

B) Only J change

C) Only K change

D) Never

Q.22 [2 Marks] [MODERATE] [EMFT — Transmission Lines]

Binary 1011 to Gray code:

A) 1011
B) 1110
C) 1101
D) 0111

Q.23 [2 Marks] [MODERATE] [Communication — AM/FM]

Polarization in EM waves refers to:

A) Electric field direction
B) Magnetic field direction
C) Wave propagation direction
D) Frequency variation

Q.24 [2 Marks] [MODERATE] [Microprocessors — 8085/8086]

Nyquist rate for signal with max frequency 5kHz:

A) 5 kHz
B) 10 kHz
C) 2.5 kHz
D) 15 kHz

Q.25 [2 Marks] [MODERATE] [Network Theory — AC Circuits]

Thevenin equivalent replaces any linear network by:

A) V-source + Resistor
B) I-source + Resistor
C) V-source + Inductor
D) I-source + Capacitor

Q.26 [2 Marks] [MODERATE] [Signals & Systems — Convolution]

Impedance of inductor L in s-domain:

A) 1/s
B) sL
C) 1/(sL)
D) s/L

Q.27 [2 Marks] [MODERATE] [Control Systems — Block Diagrams]

Routh-Hurwitz determines:

A) Steady-state error
B) RHP pole count
C) Damping ratio
D) Natural frequency

Q.28 [2 Marks] [MODERATE] [Digital Electronics — Sequential Circuits]

BJT in active region:

A) Both junctions forward

B) Both reverse

C) EB forward, CB reverse

D) EB reverse, CB forward

Q.29 [2 Marks] [MODERATE] [Analog Electronics — BJT Amplifiers]

8085 microprocessor address lines:

A) 8
B) 16
C) 20
D) 32

Q.30 [2 Marks] [MODERATE] [EMFT — Waveguides]

FM modulation index ;" v—F, ”f=75kHz, fm=15kHz:

A) 3
B) 5
C) 10
D) 15

Q.31 [2 Marks] [MODERATE] [Communication — Information Theory]

Intrinsic impedance of free space ;r $ƒ

A) 50:•
B) 120< ©
C) 377:•
D) Both B and C

Q.32 [2 Marks] [MODERATE] [Microprocessors — Interfacing]

A 4-bit DAC with Vref=5V has resolution:

A) 0.3125V
B) 0.5V
C) 1V
D) 0.25V

Q.33 [2 Marks] [HARD] [Network Theory — AC Circuits]

A JK flip-flop with J=K=1 toggles on:

A) Every clock
B) Only J change
C) Only K change
D) Never

Q.34 [2 Marks] [HARD] [Signals & Systems — LTI Systems]

Binary 1011 to Gray code:

A) 1011
B) 1110
C) 1101
D) 0111

Q.35 [2 Marks] [HARD] [Control Systems — Root Locus]

Polarization in EM waves refers to:

A) Electric field direction

B) Magnetic field direction

C) Wave propagation direction

D) Frequency variation

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Digital Electronics — Sequential Circuits]

Which are closed-loop control components?

A) Controller
B) Comparator
C) Plant
D) Feedback path
E) Reference input

Q.37 [2 Marks] [HARD] [Analog Electronics — Diodes]

Which improve steady-state error in control?

A) Increase Kp
B) Add integral
C) Add derivative
D) Add lag
E) Add lead

Q.38 [2 Marks] [HARD] [EMFT — Transmission Lines]

Which are digital modulation schemes?

A) ASK
B) FSK
C) PSK
D) AM
E) QAM

Q.39 [2 Marks] [HARD] [Communication — AM/FM]

Which are statically indeterminate structures?

A) Fixed-fixed beam
B) Cantilever
C) Continuous beam
D) Simply supported
E) Propped cantilever

Q.40 [2 Marks] [HARD] [Microprocessors — Programming]

Which are closed-loop control components?

A) Controller
B) Comparator
C) Plant
D) Feedback path
E) Reference input

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Control Systems — Bode/Nyquist]

G(s) = K/(s(s+2)). K for marginal stability?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Digital Electronics — Combinational Circuits]

CE amplifier ;#Ó Â &3ÓV³©, Re=1k:'à |Av| "H ?

Answer: ____________________

Q.53 [2 Marks] [HARD] [Analog Electronics — Diodes]

Parallel plate capacitor A=0.1m², d=1mm. C (pF)?

Answer: ____________________

Q.54 [2 Marks] [HARD] [EMFT — Electrostatics]

8085 max memory (KB)?

Answer: ____________________

Q.55 [2 Marks] [HARD] [Communication — Digital Modulation]

A 3dB bandwidth filter: |H(j<'—Â Ö , Ò à At 3dB, |H(j<'—Â Ò ð

Answer: ____________________

Q.56 [2 Marks] [HARD] [Microprocessors — Programming]

Digital gate: AND of 4 inputs, each 5V logic. Output (V)?

Answer: ____________________

Q.57 [2 Marks] [HARD] [Network Theory — AC Circuits]

3-phase load: 10A/phase, 0.8pf, 415V. Total real power (W)?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Signals & Systems — Z-transform]

For x(t) = e^(-2t)u(t), ROC is Re(s) > ?:

Answer: ________________________

Q.59 [2 Marks] [HARD] [Control Systems — Bode/Nyquist]

G(s) = K/(s(s+2)). K for marginal stability?

Answer: ____________________

Q.60 [2 Marks] [HARD] [Digital Electronics — Sequential Circuits]

CE amplifier ;#Ó Â &3ÓV³©, Re=1k:'à |Av| "H ?

Answer: ____________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	A	**Q.2:**	B	**Q.3:**	B
Q.4:	C	**Q.5:**	B	**Q.6:**	B
Q.7:	D	**Q.8:**	A	**Q.9:**	A
Q.10:	B	**Q.11:**	A	**Q.12:**	B
Q.13:	A	**Q.14:**	B	**Q.15:**	B
Q.16:	C	**Q.17:**	B	**Q.18:**	B
Q.19:	D	**Q.20:**	A	**Q.21:**	A
Q.22:	B	**Q.23:**	A	**Q.24:**	B
Q.25:	A	**Q.26:**	B	**Q.27:**	B
Q.28:	C	**Q.29:**	B	**Q.30:**	B
Q.31:	D	**Q.32:**	A	**Q.33:**	A
Q.34:	B	**Q.35:**	A	**Q.36:**	ABCDE
Q.37:	ABD	**Q.38:**	ABCE	**Q.39:**	ACE
Q.40:	ABCDE	**Q.41:**	ABD	**Q.42:**	ABCE
Q.43:	ACE	**Q.44:**	ABCDE	**Q.45:**	ABD
Q.46:	ABCE	**Q.47:**	ACE	**Q.48:**	ABCDE
Q.49:	ABD	**Q.50:**	ABCE	**Q.51:**	4
Q.52:	500	**Q.53:**	885	**Q.54:**	64
Q.55:	0.707	**Q.56:**	5	**Q.57:**	5744
Q.58:	2	**Q.59:**	4	**Q.60:**	500

Detailed Solutions

Question 1
[Network Theory — AC Circuits]

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: Vth in series with Rth. Norton: In in parallel with Rn.

Question 2
[Signals & Systems — Convolution]

Impedance of inductor L in s-domain:

Answer: (B)

Z_L(s) = sL (voltage leads current by 90°).

Question 3
[Control Systems — Block Diagrams]

Routh-Hurwitz determines:

Answer: (B)

Counts poles in Right Half-Plane to assess stability.

Question 4
[Digital Electronics — Combinational Circuits]

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 5
[Analog Electronics — Feedback]

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !' 64KB memory addressing.

Question 6
[EMFT — Magnetostatics]

FM modulation index ;" v—F, "f=75kHz, fm=15kHz:

Answer: (B)

;" Ò "f/fm = 75/15 = 5.

Question 7
[Communication — Digital Modulation]

Intrinsic impedance of free space ;r $ƒ

Answer: (D)

;r Ò !¢ƒ¼ €/;R ' $, 3ss© = 120< ©.

Question 8
[Microprocessors — Programming]

A 4-bit DAC with Vref=5V has resolution:

Answer: (A)

Resolution = 5V/16 = 0.3125V.

Question 9
[Network Theory — Thevenin/Norton]

A JK flip-flop with J=K=1 toggles on:

Answer:

(A)

J=K=1 is toggle mode. Output inverts each clock.

Question 10 *[Signals & Systems — LTI Systems]*

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits !’ 1110.

Question 11 *[Control Systems — Routh-Hurwitz]*

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 12 *[Digital Electronics — Boolean Algebra]*

Nyquist rate for signal with max frequency 5kHz:

Answer: (B)

Nyquist rate = 2 × f_max = 2 × 5 = 10 kHz.

Question 13 *[Analog Electronics — Feedback]*

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: Vth in series with Rth. Norton: In in parallel with Rn.

Question 14 *[EMFT — Transmission Lines]*

Impedance of inductor L in s-domain:

Answer: (B)

Z_L(s) = sL (voltage leads current by 90°).

Question 15 *[Communication — AM/FM]*

Routh-Hurwitz determines:

Answer: (B)

Counts poles in Right Half-Plane to assess stability.

Question 16 *[Microprocessors — 8085/8086]*

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 17 *[Network Theory — Three-phase]*

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !’ 64KB memory addressing.

Question 18 *[Signals & Systems — Laplace Transform]*

FM modulation index ;" v—F, "f=75kHz, fm=15kHz:

Answer:

(B)

;" Ò "f/fm = 75/15 = 5.

Question 19 *[Control Systems — Block Diagrams]*

Intrinsic impedance of free space ;r $f

Answer: (D)

;r Ò !¢ƒ¼ €/;R ' $, 3ss© = 120< ©.

Question 20 *[Digital Electronics — Combinational Circuits]*

A 4-bit DAC with Vref=5V has resolution:

Answer: (A)

Resolution = 5V/16 = 0.3125V.

Question 21 *[Analog Electronics — Diodes]*

A JK flip-flop with J=K=1 toggles on:

Answer: (A)

J=K=1 is toggle mode. Output inverts each clock.

Question 22 *[EMFT — Transmission Lines]*

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits !' 1110.

Question 23 *[Communication — AM/FM]*

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 24 *[Microprocessors — 8085/8086]*

Nyquist rate for signal with max frequency 5kHz:

Answer: (B)

Nyquist rate = 2 × f_max = 2 × 5 = 10 kHz.

Question 25 *[Network Theory — AC Circuits]*

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: Vth in series with Rth. Norton: In in parallel with Rn.

Question 26 *[Signals & Systems — Convolution]*

Impedance of inductor L in s-domain:

Answer: (B)

Z_L(s) = sL (voltage leads current by 90°).

Question 27 *[Control Systems — Block Diagrams]*

Routh-Hurwitz determines:

Answer:

(B)

Counts poles in Right Half-Plane to assess stability.

Question 28 *[Digital Electronics — Sequential Circuits]*

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 29 *[Analog Electronics — BJT Amplifiers]*

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !’ 64KB memory addressing.

Question 30 *[EMFT — Waveguides]*

FM modulation index ;" v—F, ”f=75kHz, fm=15kHz:

Answer: (B)

;" Ò ”f/fm = 75/15 = 5.

Question 31 *[Communication — Information Theory]*

Intrinsic impedance of free space ;r $f

Answer: (D)

;r Ò !¢ƒ¼ €/;R ’ $, 3ss© = 120< ©.

Question 32 *[Microprocessors — Interfacing]*

A 4-bit DAC with Vref=5V has resolution:

Answer: (A)

Resolution = 5V/16 = 0.3125V.

Question 33 *[Network Theory — AC Circuits]*

A JK flip-flop with J=K=1 toggles on:

Answer: (A)

J=K=1 is toggle mode. Output inverts each clock.

Question 34 *[Signals & Systems — LTI Systems]*

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits !’ 1110.

Question 35 *[Control Systems — Root Locus]*

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 36 *[Digital Electronics — Sequential Circuits]*

Which are closed-loop control components?

Answer:

(A, B, C, D, E)

All listed are components of a feedback control loop.

Question 37 *[Analog Electronics — Diodes]*

Which improve steady-state error in control?

Answer: (A, B, D)

Kp, integral action, lag compensator reduce SSE.

Question 38 *[EMFT — Transmission Lines]*

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 39 *[Communication — AM/FM]*

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 40 *[Microprocessors — Programming]*

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 41 *[Network Theory — Three-phase]*

Which improve steady-state error in control?

Answer: (A, B, D)

Kp, integral action, lag compensator reduce SSE.

Question 42 *[Signals & Systems — Fourier Series]*

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 43 *[Control Systems — Bode/Nyquist]*

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 44 *[Digital Electronics — Combinational Circuits]*

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 45 *[Analog Electronics — Feedback]*

Which improve steady-state error in control?

Answer:

(A, B, D)

Kp, integral action, lag compensator reduce SSE.

Question 46
[EMFT — Transmission Lines]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 47
[Communication — Digital Modulation]

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 48
[Microprocessors — 8085/8086]

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 49
[Network Theory — AC Circuits]

Which improve steady-state error in control?

Answer: (A, B, D)

Kp, integral action, lag compensator reduce SSE.

Question 50
[Signals & Systems — Laplace Transform]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 51
[Control Systems — Bode/Nyquist]

G(s) = K/(s(s+2)). K for marginal stability?

Answer: (4)

$s^2 + 2s + K = 0$. Marginal stability when $K = 2 \times 2 = 4$.

Question 52
[Digital Electronics — Combinational Circuits]

CE amplifier ;#Ó Â &3ÓV³©, Re=1k:'à |Av| "H ?

Answer: (500)

Av "H -;%&2õ&R Ò Ó sS ó Ò ÓS à |Av| = 500.

Question 53
[Analog Electronics — Diodes]

Parallel plate capacitor A=0.1m², d=1mm. C (pF)?

Answer: (885)

C = ;R öB Ò ,ãƒ]s »›-s ã ó ã Ò ƒƒR `.

Question 54
[EMFT — Electrostatics]

8085 max memory (KB)?

Answer:

(64)

16-bit address bus !' 2^16 = 65536 bytes = 64 KB.

Question 55 *[Communication — Digital Modulation]*

A 3dB bandwidth filter: |H(j<'—Â Ö , Ò à At 3dB, |H(j<'—Â Ò ð

Answer: (0.707)

3dB point = 1/" 2 "H 0.707 of maximum response.

Question 56 *[Microprocessors — Programming]*

Digital gate: AND of 4 inputs, each 5V logic. Output (V)?

Answer: (5)

AND gate outputs HIGH (5V) when all inputs are HIGH.

Question 57 *[Network Theory — AC Circuits]*

3-phase load: 10A/phase, 0.8pf, 415V. Total real power (W)?

Answer: (5744)

P = " 3 × V_L × I_L × cos<b Ò ãs3" r C R r   r ã, Ò SsCB W.

Question 58 *[Signals & Systems — Z-transform]*

For x(t) = e^(-2t)u(t), ROC is Re(s) > ?.

Answer: (2)

L{e^(-at)u(t)} has ROC: Re(s) > a. Here a=2.

Question 59 *[Control Systems — Bode/Nyquist]*

G(s) = K/(s(s+2)). K for marginal stability?

Answer: (4)

$s^2 + 2s + K = 0$. Marginal stability when K = 2×2 = 4.

Question 60 *[Digital Electronics — Sequential Circuits]*

CE amplifier ;#Ó   Â &3ÓV³©, Re=1k:'à |Av| "H ?

Answer: (500)

Av "H -;%&2õ&R Ò Ó   sS   ó   Ò ÓS à |Av| = 500.
